

DATASET DOCUMENTATION

CYCLISTIC BIKE-SHARE DATA ANALYSIS PROJECT

This documentation serves as a reference throughout the data preparation, cleaning, transformation, and analysis phases, helping maintain consistency and accuracy in the interpretation of the data.

Dataset Overview

The dataset consists of Cyclicistic's historical bike-trip records collected over the previous 12 months. Each record represents a single completed bike trip and contains information about ride details, station locations, timestamps, and rider membership type. The dataset is provided in monthly CSV files and was consolidated into a single dataset for analysis.

Data Source

The following table summarizes the primary source of the dataset used in this project, including information about the dataset name, provider, owner, format, license and geographic coverage. This information provides context for understanding the origin and scope of the data used throughout the analysis.

Tabel 1. Information about dataset

Attribute	Information
Dataset Name	Divvy Bike Share Historical Trip Data
Dataset Owner	The City of Chicago
Dataset Provider	Divvy Bike Share (operated by Lyft Bikes and Scooters, LLC.)
Location	Chicago, Illinois, US
Data Format	CSV
License	Publicly available under the Divvy Data License Agreement

Data Structure

The Cyclistic historical bike-trip dataset is organized into multiple monthly CSV files, with each record representing a single completed bike trip. The table below summarizes the overall structure of the dataset, including its format, observation unit, geographic coverage, and key characteristics. Understanding the dataset structure helps ensure that the data is interpreted correctly before proceeding with the cleaning, transformation, and analysis phases.

Tabel 2. Structure of the dataset

Attribute	Description
Files Name	[year][month]-divvy-tripdata
Number of Files	12 monthly CSV files
Data Format	CSV
Time Period	July 2025 – June 2026
Observation Unit	One completed bike trip
Geographic Coverage	Chicago, Illinois, USA
Primary Identifier	ride_id
Target Variable	Member_casual
Number of Variables	13 original variables

Data Dictionary

To ensure consistency and accuracy throughout the analysis, a data dictionary was created to document each variable in the Cyclistic historical bike-trip dataset. The dictionary includes the column name, data type, and a brief description of each variable, providing a standardized reference for data preparation, transformation, and analysis.

Tabel 3. Type and description of variable in dataset

Column	Type	Description
ride_id	String	ID attached to each trip taken
rideable_type	String	Type of bike used
started_at	String	Day and time trip started, in CST

Column	Type	Description
ended_at	String	Day and time trip ended, in CST
start_station_name	String	Name of station where trip originated
start_station_id	String	ID of station where trip originated
end_station_name	String	Name of station where trip terminated
end_station_id	String	ID of station where trip terminated
start_lat	Float64	Latitude coordinate of station where trip originated
start_lng	Float64	Longitude coordinate of station where trip originated
end_lat	Float64	Latitude coordinate of station where trip terminated
end_lng	Float64	Longitude coordinate of station where trip terminated
member_casual	String	Type of rider based on purchased

The **rideable_type** variable identifies the type of bicycle used for each trip. Understanding the available ride types is important for analyzing rider preferences and comparing bike usage patterns between annual members and casual riders. The table below describes each value found in the dataset.

Tabel 4. Value of variable rideable_type description

Category	Description
classic_bike	Traditional pedal-powered bicycle
electric_bike	Bicycle equipped with an electric motor that assists the rider while pedaling.

The **member_casual** variable identifies the membership status of each rider and serves as the primary variable for customer segmentation in this analysis. It distinguishes between annual members and casual riders, enabling comparisons of their riding behaviors, usage patterns, and travel preferences. The table below describes each membership category included in the dataset.

Tabel 5. Value of variable member_casual description

Category	Description
casual	Rider who purchased a 24-Hour Pass

Category	Description
member	Rider who purchased an Annual Membership

Data Limitation

Although the Cyclistic historical bike-trip dataset provides comprehensive information about bike usage, it has several limitations that should be considered during the analysis. The dataset does not include personally identifiable information or customer demographics such as age, gender, or income, making it impossible to analyze rider behavior based on these characteristics.

Key Limitations

- The dataset does not include customer demographic information.
- Personally identifiable information (PII) has been removed to protect user privacy.
- The purpose or motivation of each trip is not recorded.
- External factors such as weather, traffic conditions, and local events are not included.
- The analysis is limited to historical trip records and may not reflect future rider behavior

Data Storage

The Cyclistic historical bike-trip dataset was downloaded as monthly CSV files from the Divvy Bike Share public data repository. All files were stored locally in a dedicated project folder and organized by month to maintain a consistent file structure. Before analysis, the monthly datasets were combined into a single master dataset and securely maintained throughout the data preparation and analysis process. The processed dataset was then used for data cleaning, transformation, exploratory analysis, and visualization.

Table 6. Storage detail of dataset

Attribute	Information
Storage Format	CSV
Storage Location	Local project folder
Analysis Tool	Python (Pandas) in Jupyter Notebook
Processed Data	Saved as a cleaner CSV file for subsequent analysis